

CASE STUDY

Statistical Arbitrage Portfolio Construction and Hypothetical Country Valuation

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Google Drive – Complete Project Files:

<https://drive.google.com/drive/folders/19QsBfTYreG8TYhiq8ADTlzdQw7oHiOWQ?usp=sharing>

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EXECUTIVE SUMMARY

This case study consists of two independent quantitative problems. The first problem focuses on the identification, construction, validation, and backtesting of a mean-reverting statistical arbitrage portfolio. The second problem develops a hypothetical framework for valuing nation-states as if they could be bought, sold, and held as economic assets.

The objective of the first analysis was not simply to identify two highly correlated assets, but to determine whether an economically related pair exhibited genuine statistical mean reversion. The research initially examined Gold and Silver because both precious metals are influenced by similar macroeconomic factors. The Gold-Silver data showed a very high price correlation of approximately 0.9654. However, the Engle-Granger cointegration test produced a p-value of 0.177788, which was not statistically significant at the 5% level. Therefore, the Gold-Silver pair was not accepted as the final statistical arbitrage portfolio.

The analysis then considered the Energy Select Sector SPDR ETF (XLE) and the Vanguard Energy ETF (VDE). These ETFs are economically linked because both provide exposure to the United States energy sector. The pair showed an extremely high correlation of approximately 0.9995, and the Engle-Granger cointegration test produced a p-value of 0.008646. A hedge ratio was estimated using ordinary least squares regression, and a spread was constructed and tested for stationarity. The strategy was then tested using a 70% training period and a 30% out-of-sample period. The Augmented Dickey-Fuller test provided evidence of stationarity in both periods.

The trading strategy used a 60-day rolling Z-score to identify deviations of the spread from its historical mean. Positions were entered when the Z-score reached extreme levels and were closed when the spread reverted towards its mean. Although the statistical tests supported the mean-reversion hypothesis, the final backtest produced negative returns, particularly after transaction costs. This demonstrates an important distinction between a statistically significant relationship and a practically profitable investment strategy.

In the second analysis, a quantitative framework was developed to value countries hypothetically as economic assets. Five countries with different economic sizes and resource profiles were selected: the United States, China, Saudi Arabia, India, and Singapore. The model used two factors: Gross Domestic Product and total natural resource rents as a percentage of GDP. GDP was assigned a weight of 70%, while natural resources were assigned a weight of 30%. All variables were normalized to a 0–100 scale before calculating the final Country Valuation Score. Based on 2021 data, the United States ranked first with a score of 71.50, followed by China with 55.71, Saudi Arabia with 32.90, India with 13.05, and Singapore with 1.30.

Overall, the case study demonstrates the importance of combining data analysis, statistical testing, economic reasoning, out-of-sample validation, realistic assumptions, and self-critique. A statistically significant model is not automatically profitable, and a numerical valuation model is only as reliable as its assumptions and the variables used.

1. INTRODUCTION

Quantitative analysis plays an important role in modern investment decision-making. However, a useful quantitative model requires more than producing attractive charts or identifying high correlations. The analyst must select appropriate data, justify the economic relationship between variables, test statistical assumptions, avoid look-ahead bias, validate results using unseen data, and honestly identify the limitations of the model.

This case study applies this approach to two different problems.

The first problem investigates whether a pair of financially related assets can be used to construct a mean-reverting statistical arbitrage portfolio. Statistical arbitrage strategies attempt to identify temporary deviations between related securities and profit when the relationship returns towards its historical equilibrium.

The second problem applies a similar quantitative mindset to a more unusual question. If countries could hypothetically be treated as assets that could be bought and held on a balance sheet, how could they be valued? Instead of providing a geopolitical discussion, the objective was to construct a numerical model using real economic data and produce a transparent ranking.

The two questions are different in subject matter, but both require the same core analytical principles:

- Selection of appropriate variables
- Reliable data collection
- Data cleaning and preparation
- Statistical testing
- Quantitative modeling
- Validation of assumptions
- Numerical interpretation of results
- Identification of model limitations

The case study instructions required a written report, reproducible code, and raw or derived data, with the final work expected to produce defensible and numerically supported conclusions. The evaluation also emphasizes statistical rigor, modeling judgment, data handling, code quality, communication, and self-critique.

2. OBJECTIVES OF THE STUDY

2.1 Objective of Question 1

The objective of Question 1 was to identify, construct, and validate a mean-reverting portfolio using economically related financial assets.

The specific objectives were:

1. To identify economically related asset pairs.
2. To collect historical price data.
3. To clean and align the datasets.
4. To examine correlation between the selected assets.
5. To test for cointegration using the Engle-Granger approach.
6. To estimate a hedge ratio.
7. To construct a trading spread.
8. To test the spread for stationarity using the Augmented Dickey-Fuller test.
9. To construct a trading strategy using Z-scores.
10. To perform a backtest.
11. To include transaction costs.
12. To validate the model using an out-of-sample test.
13. To determine whether statistical mean reversion translated into practical investment profitability.

2.2 Objective of Question 2

The objective of Question 2 was to develop a quantitative framework for hypothetically valuing countries as if they were investable economic assets.

The specific objectives were:

1. To select real countries with different economic and resource characteristics.
2. To collect real economic data.
3. To define measurable valuation factors.
4. To normalize variables so they could be compared.
5. To assign transparent weights to the selected factors.
6. To calculate a numerical Country Valuation Score.
7. To rank the selected countries.
8. To interpret the results.
9. To identify the assumptions and limitations of the valuation framework.

PART A: QUESTION 1

3. MEAN-REVERSION AND STATISTICAL ARBITRAGE ANALYSIS

3.1 Introduction to Mean Reversion

Mean reversion is based on the idea that a variable that moves significantly away from its long-term average may eventually move back towards its normal or equilibrium level.

In a pairs trading strategy, two economically related assets are analysed together. Instead of predicting whether the entire market will rise or fall, the strategy focuses on the relative relationship between the two assets.

If the relationship between the assets temporarily moves away from its historical equilibrium, a trader may take opposite positions in the two securities. The expectation is that the spread between them will eventually revert towards its historical mean.

However, high correlation alone is not sufficient evidence of a valid mean-reverting relationship. Two assets can move together for a long period while their price difference continues to drift without returning to a stable equilibrium. Therefore, statistical tests such as cointegration and stationarity tests are required.

4. DATA COLLECTION AND PREPARATION

Historical market data was collected using Yahoo Finance data.

The analysis initially considered Gold and Silver because they are economically related precious metals. Both assets are influenced by factors such as inflation expectations, interest rates, currency movements, investor demand for safe-haven assets, and commodity market conditions.

The common dataset contained approximately 1,120 aligned observations after data cleaning. The final common period used in the analysis covered approximately February 2022 to August 2026. Missing values were checked and rows without both asset prices were removed. No missing values remained in the final aligned Gold-Silver dataset.

The following data preparation steps were performed:

1. Conversion of the Date column into datetime format.
2. Sorting of observations chronologically.

3. Identification of the common date range.
4. Alignment of both assets by date.
5. Calculation of daily returns.
6. Removal of missing observations.
7. Calculation of cumulative returns.
8. Calculation of volatility.
9. Calculation of drawdowns.
10. Preparation of the final dataset for further statistical analysis.

The Gold and Silver datasets contained the same number of aligned observations after preparation, allowing a direct comparison between the two assets.

5. EXPLORATORY ANALYSIS OF GOLD AND SILVER

5.1 Correlation Analysis

The first stage of the analysis examined the relationship between Gold and Silver prices.

The calculated Gold-Silver price correlation was:

Gold-Silver Correlation = 0.9654

This indicates a very strong positive relationship between the two price series. Gold and Silver generally moved in the same direction during the selected period.

However, correlation does not prove that the relationship is stable enough for a statistical arbitrage strategy. A high correlation may simply indicate that both assets are responding to similar market conditions.

Therefore, additional statistical testing was required.

Price Trend



The trends of Adj Close Gold and Adj Close Silver for Date. Details are shown for Adj Close Gold and Adj Close Silver.

Figure 1: Gold vs Silver Price Trend

Insert your Gold vs Silver price chart here.

5.2 Gold and Silver Return Comparison

The performance comparison showed that both Gold and Silver produced substantial positive cumulative returns over the selected period.

The results were:

Asset	Total Return
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Gold ETF	203.92%
Silver ETF	263.55%

A hypothetical investment of ₹100,000 would have grown approximately to:

Investment	Final Value
Gold	₹303,918.28
Silver	₹363,552.25

Silver produced a higher total return during the analysed period, but it also experienced greater volatility and a larger maximum drawdown.

The annualized volatility results were approximately:

- Gold: 18.91%
- Silver: 34.34%

The maximum drawdown results were approximately:

- Gold: -24.44%
- Silver: -45.01%

This indicates that Silver generated a higher return but was also associated with substantially higher risk.

Cumulative Return



The trends of Cumulative Return Gold and Cumulative Return Silver for Date. Details are shown for Cumulative Return Gold and Cumulative Return Silver.

Figure 2: Gold vs Silver Cumulative Return

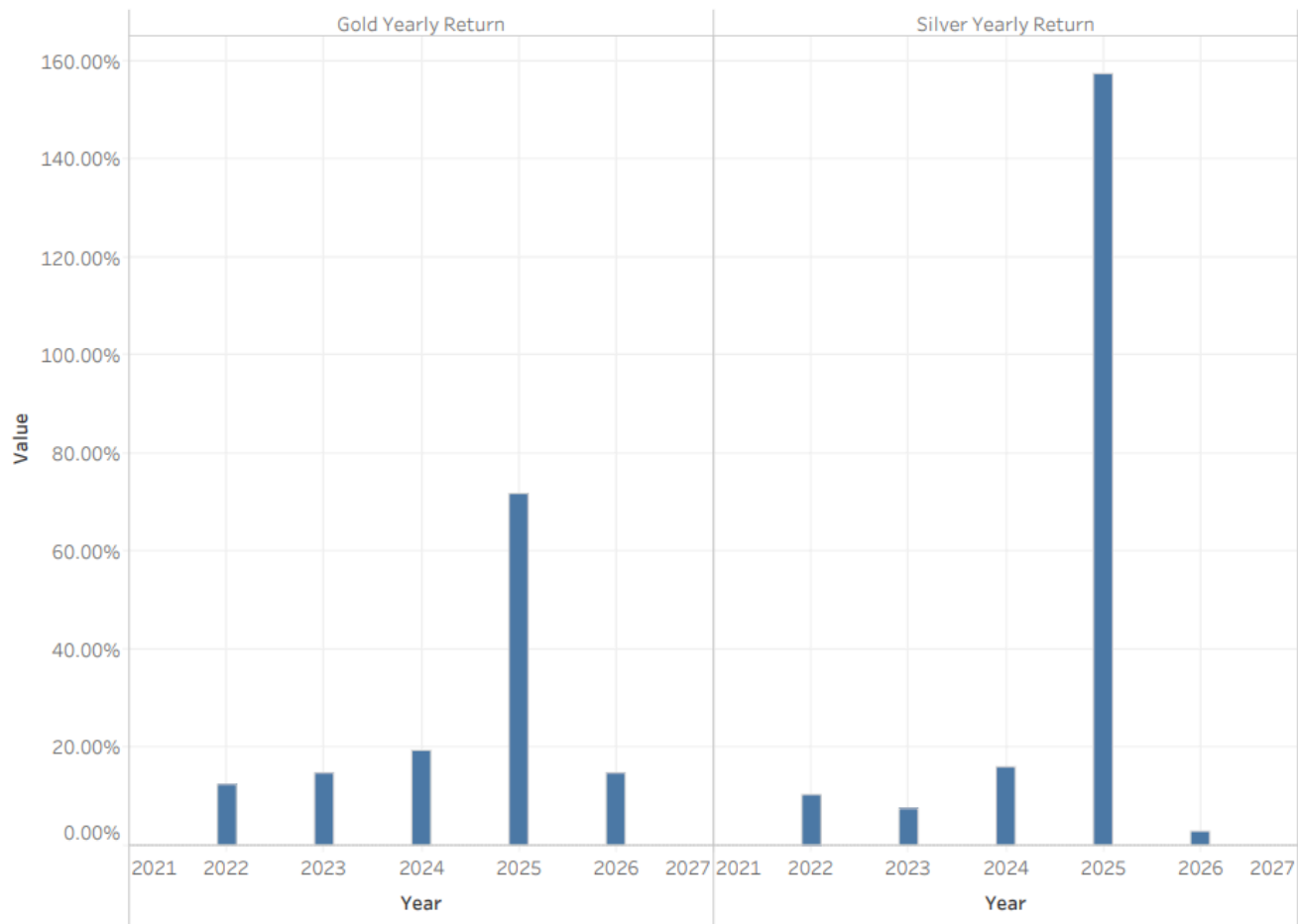
₹1 Lakh Investment Growth: Gold vs Silver



The trends of Investment Value Gold and Investment Value Silver for Date. Details are shown for Investment Value Gold and Investment Value Silver.

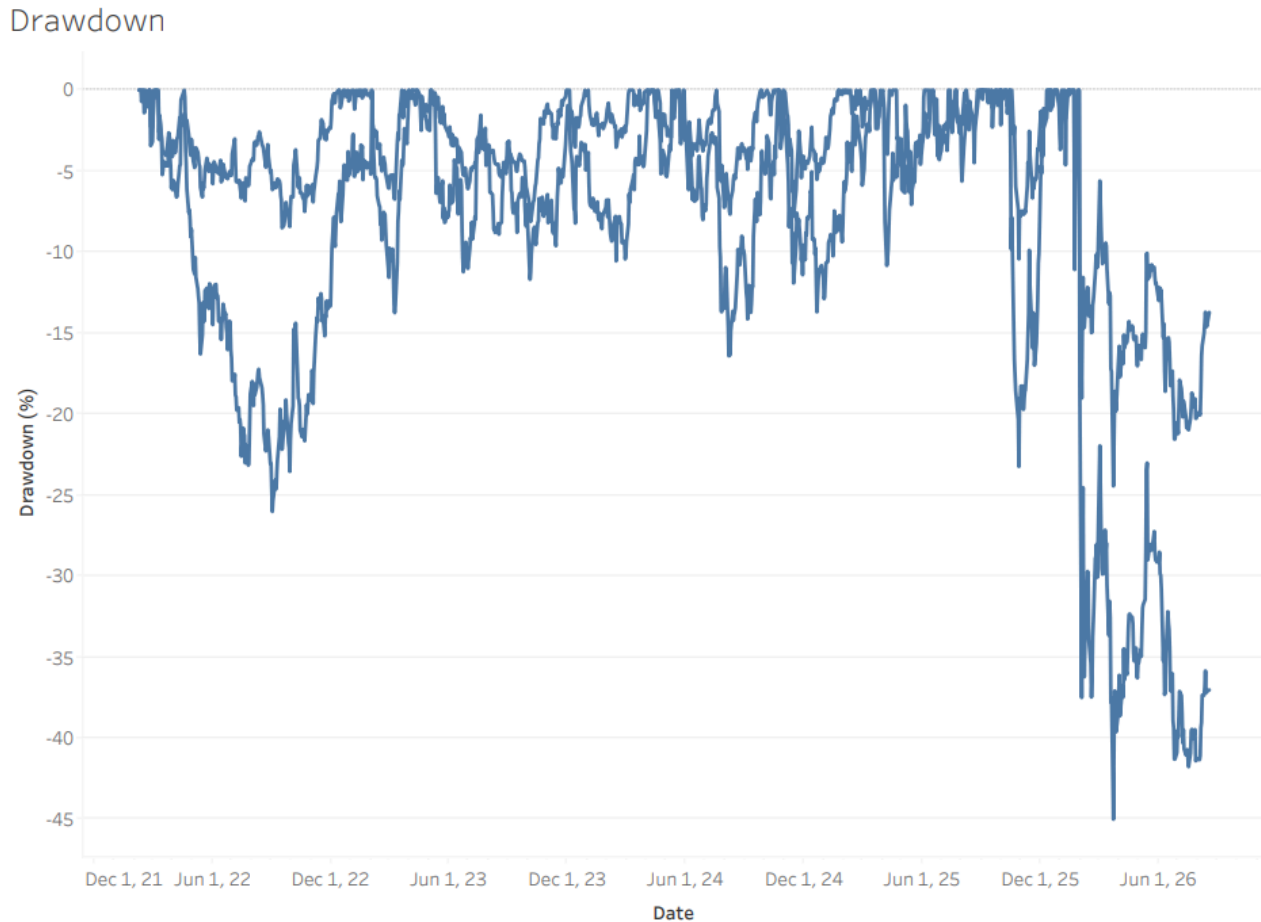
Figure 3: ₹1 Lakh Investment Growth

Gold vs Silver — Annual Returns



The plots of Gold Yearly Return and Silver Yearly Return for Year. Details are shown for Gold Yearly Return and Silver Yearly Return.

Figure 4: Gold vs Silver Annual Returns



The trends of Drawdown Gold and Drawdown Silver for Date. Details are shown for Drawdown Gold and Drawdown Silver.

Figure 5: Gold vs Silver Drawdown

You can take these screenshots from your Tableau dashboard.

6. GOLD-SILVER COINTEGRATION TEST

The Engle-Granger cointegration test was performed to determine whether Gold and Silver had a stable long-term equilibrium relationship.

The results were:

- Test Statistic: -2.7629
- P-value: 0.177788

The 5% critical value was approximately -3.3416.

Since the p-value was greater than 0.05, the null hypothesis of no cointegration could not be rejected at the conventional 5% significance level.

Therefore, despite the very high correlation between Gold and Silver, the evidence was not sufficiently strong to classify the pair as a statistically validated cointegrated portfolio under the selected test and sample.

This result is important because it demonstrates why correlation alone should not be used to select pairs for statistical arbitrage.

Instead of forcing a Gold-Silver trading strategy, the analysis proceeded to another economically linked pair.

7. SELECTION OF THE FINAL PORTFOLIO: XLE AND VDE

The final portfolio consisted of:

- **XLE – Energy Select Sector SPDR ETF**
- **VDE – Vanguard Energy ETF**

These ETFs were selected because they have a strong economic relationship. Both provide exposure to the United States energy sector and are influenced by many of the same underlying industry and market factors.

This provides an economic rationale for testing the pair.

Historical data for XLE and VDE was downloaded and aligned. The analysis covered the period from approximately January 2022 to August 2026.

The calculated correlation was:

XLE-VDE Correlation = 0.9995

The pair therefore showed an extremely strong historical relationship. However, as discussed earlier, correlation alone was not considered sufficient. The pair was also tested for cointegration.

8. COINTEGRATION ANALYSIS OF XLE AND VDE

The Engle-Granger cointegration test was performed on the XLE and VDE price series.

The results were:

- Test Statistic: -3.9429
- P-value: 0.008646

Because the p-value was below 0.05, the null hypothesis of no cointegration was rejected at the 5% significance level. This provided statistical evidence of a long-term equilibrium relationship between the two ETFs during the analysed sample.

This result was stronger than the Gold-Silver result and therefore XLE and VDE were selected as the final candidate portfolio for the mean-reversion analysis.

9. HEDGE RATIO ESTIMATION

A hedge ratio was estimated using Ordinary Least Squares regression.

The basic relationship was expressed as:

$$\text{XLE} = \text{Alpha} + \text{Beta} \times \text{VDE} + \text{Error}$$

The estimated hedge ratio represents the amount of VDE required to hedge the relative exposure of XLE.

The analysis estimated a hedge ratio of approximately:

$$\text{Beta} = 0.348405$$

The regression also produced:

$$\text{R-squared} = 0.999053$$

This indicates that, during the analysed full sample, the two ETFs had an extremely close linear relationship.

The hedge ratio was then used to construct the spread:

$$\text{Spread} = \text{XLE} - (\text{Hedge Ratio} \times \text{VDE})$$

The spread represents the relative deviation between the two assets after adjusting for their different price levels.

A stationary spread is important because mean-reversion trading depends on the expectation that unusually large deviations will eventually move back towards their historical equilibrium.

10. TRAINING AND OUT-OF-SAMPLE VALIDATION

One of the most important parts of the analysis was the separation of the data into training and testing periods.

The data was divided approximately as follows:

- Training Data: 70%
- Out-of-Sample Testing Data: 30%

The training dataset contained 770 observations, while the test dataset contained 331 observations.

The approximate periods were:

- Training Period: 29 March 2022 to 23 April 2025
- Testing Period: 24 April 2025 to 18 August 2026

This separation was important because a model that performs well only on the same data used to construct it may be overfitted.

The hedge ratio was estimated using the training data. The same estimated relationship was then applied to the unseen out-of-sample data.

11. TRAINING PERIOD ADF TEST

The Augmented Dickey-Fuller test was performed on the training-period spread.

The results were:

Measure	Result
Hedge Ratio	0.353302
ADF Statistic	-3.062554
P-value	0.029451
1% Critical Value	-3.438984
5% Critical Value	-2.865350
10% Critical Value	-2.568799

Since the p-value of 0.029451 was below 0.05, the null hypothesis of a unit root was rejected at the 5% significance level.

This provides evidence that the training-period spread was stationary and therefore consistent with the mean-reversion hypothesis.

12. OUT-OF-SAMPLE ADF TEST

The most important robustness check was performed on the unseen testing period.

The hedge ratio estimated from the training data was used without re-estimating the model on the test data.

The out-of-sample results were:

Measure	Result
ADF Statistic	-3.117642
P-value	0.025275
1% Critical Value	-3.450384
5% Critical Value	-2.870365
10% Critical Value	-2.571472

The p-value of 0.025275 was below 0.05.

Therefore, the null hypothesis of a unit root was also rejected during the out-of-sample period. This provided additional evidence that the spread remained stationary in data that was not used to estimate the hedge ratio.

This was an important result because it reduced the risk that the observed mean reversion was purely the result of fitting the model to historical data.

13. Z-SCORE TRADING STRATEGY

After constructing the spread, a 60-day rolling window was used to calculate:

1. Rolling Mean
2. Rolling Standard Deviation
3. Z-Score

The Z-score was calculated as:

$$\text{Z-Score} = (\text{Current Spread} - \text{Rolling Mean}) / \text{Rolling Standard Deviation}$$

The Z-score measures how far the current spread is from its recent average in terms of standard deviations.

A 60-day rolling window was used in the analysis.

The trading rules were defined as follows:

Long Spread

When the Z-score was less than or equal to -2:

- Go long the relatively undervalued side of the spread.
- Take an offsetting short position in the other ETF.

Short Spread

When the Z-score was greater than or equal to +2:

- Go short the relatively overvalued side of the spread.
- Take an offsetting long position in the other ETF.

Exit Rule

The position was closed when the spread moved back towards its mean and the Z-score returned to approximately zero.

The use of extreme entry thresholds was intended to reduce unnecessary trading and focus only on relatively large deviations.

14. BACKTEST RESULTS

The strategy was backtested using the historical data and the defined trading rules.

The weighted version of the mean-reversion strategy produced the following gross performance:

Performance Measure	Result
Total Return	-2.38%
Annualized Return	-0.55%
Annualized Volatility	3.74%
Sharpe Ratio	-0.1473
Maximum Drawdown	-8.37%
Win Rate	44.76%

Number of Entries	45
Active Trading Days	143

These results show that although the portfolio demonstrated evidence of statistical mean reversion, the implemented trading strategy did not generate a positive gross return over the selected backtesting period.

This is one of the most important findings of the entire case study.

A statistically significant relationship does not automatically produce a profitable investment strategy.

Possible reasons include:

- Entry and exit thresholds may not be optimal.
- Mean reversion may occur too slowly.
- The spread can experience extended deviations.
- The relationship between assets can change over time.
- Trading signals may be generated at unfavorable times.
- Market microstructure and transaction costs can reduce profitability.

15. TRANSACTION COST ANALYSIS

A realistic backtest should include trading frictions.

The analysis assumed a transaction cost of:

10 Basis Points = 0.10% per transaction

Transaction costs were charged whenever the strategy position changed.

After including transaction costs, the final performance was:

Performance Measure	Result
Total Return	-10.79%
Annualized Return	-2.58%
Annualized Volatility	3.74%
Sharpe Ratio	-0.6898
Maximum Drawdown	-12.70%

Total Transaction Costs	9.00%
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The after-cost results demonstrate that trading frictions had a significant negative effect on the strategy.

This result highlights the importance of including realistic costs when evaluating a statistical arbitrage strategy. A strategy that appears potentially attractive before costs can become unattractive after frequent trading, commissions, bid-ask spreads, and other implementation costs are considered.

16. QUESTION 1 CONCLUSION

The first part of the case study investigated the construction of a mean-reverting statistical arbitrage portfolio.

The research process began with Gold and Silver because the two precious metals have a clear economic relationship and showed a high historical correlation. However, the cointegration test did not provide sufficient statistical evidence at the 5% significance level. Therefore, Gold and Silver were not accepted as the final portfolio.

The analysis then moved to XLE and VDE, two ETFs with strong economic links because both provide exposure to the United States energy sector.

The XLE-VDE pair produced strong evidence of cointegration, with an Engle-Granger p-value of 0.008646. The spread was then tested using an Augmented Dickey-Fuller test.

The training-period ADF p-value was 0.029451, while the out-of-sample ADF p-value was 0.025275. Both values were below the 5% significance level, providing evidence that the constructed spread was stationary during both the training and testing periods.

Therefore, the statistical evidence supported the mean-reversion hypothesis.

However, the backtest produced a negative return. The final weighted strategy generated a gross total return of -2.38%, and the result declined to -10.79% after including transaction costs.

The main conclusion is therefore:

The XLE-VDE portfolio demonstrated statistically significant evidence of mean reversion, but the specific trading strategy tested in this study was not practically profitable over the selected sample, especially after transaction costs.

This distinction between **statistical validity** and **investment profitability** is the most important conclusion from Question 1.

PART B: QUESTION 2

17. HYPOTHETICAL COUNTRY VALUATION MODEL

17.1 Introduction

The second part of the case study examines a hypothetical question:

If countries could be bought, sold, and held as assets, how could they be valued quantitatively?

The purpose of this analysis was not to create a geopolitical discussion or suggest that countries can actually be purchased. Political feasibility, sovereignty, legal constraints, and regulatory restrictions were intentionally set aside.

Instead, the objective was to apply the same quantitative discipline used in valuing companies or infrastructure assets.

A country can be considered, hypothetically, as a large economic system that contains productive capacity, natural resources, infrastructure, human capital, and economic activity.

For this initial model, two measurable variables were selected:

1. Economic Size, measured by GDP.
2. Natural Resource Wealth, measured by Total Natural Resource Rents as a Percentage of GDP.

The assignment required the use of real data for at least five countries with different income and resource profiles and the production of a ranked numerical output.

18. COUNTRIES SELECTED

Five countries were selected:

1. United States
2. China
3. Saudi Arabia
4. India
5. Singapore

These countries were selected because they provide different economic and resource characteristics.

United States

The United States was included because it has the largest GDP among the countries selected.

China

China was included because of its very large economic size and major role in the global economy.

India

India was included because it is a large and rapidly developing economy with a different GDP and resource profile.

Saudi Arabia

Saudi Arabia was included because natural resources, particularly the resource sector, are an important part of its economy.

Singapore

Singapore was included as an example of a highly developed economy with relatively limited natural resource rents.

The selection therefore creates variation across both economic size and resource dependence.

19. DATA SOURCES

The analysis used World Bank data for the year 2021.

The two main datasets were:

GDP Data

Indicator:

GDP (Current US Dollars)

Natural Resource Data

Indicator:

Total Natural Resources Rents (% of GDP)

The data was loaded into Python using Excel files, cleaned, filtered for the selected countries, and merged into one dataset.

The 2021 values used were:

Country	GDP 2021	Natural Resources (% of GDP)
China	18.20 Trillion USD	1.71%
India	3.17 Trillion USD	3.16%
Saudi Arabia	0.98 Trillion USD	25.57%
Singapore	0.44 Trillion USD	Approximately 0.00%
United States	23.73 Trillion USD	1.28%

These figures formed the input data for the valuation model.

20. VALUATION METHODOLOGY

The main challenge in Question 2 was that GDP and natural resource rents are measured on completely different scales.

GDP is measured in dollars, while natural resource rents are measured as a percentage of GDP.

Therefore, the variables were normalized to a common 0–100 scale.

20.1 GDP Score

The GDP score was calculated as:

$$\text{GDP Score} = (\text{Country GDP} / \text{Maximum GDP in Sample}) \times 100$$

The country with the highest GDP therefore received a score of 100.

All other countries received a score relative to the largest economy in the selected sample.

20.2 Resource Score

The Resource Score was calculated as:

$$\text{Resource Score} = (\text{Country Natural Resource Rents} / \text{Maximum Resource Rents in Sample}) \times 100$$

The country with the highest natural resource value received a score of 100.

All other countries were scored relative to that value.

21. COUNTRY VALUATION FORMULA

The final Country Valuation Score was calculated using the following weighted formula:

Country Valuation Score =

$0.70 \times \text{GDP Score} + 0.30 \times \text{Resource Score}$

The weights used were:

- GDP Weight: 70%
- Natural Resource Weight: 30%

GDP was assigned the larger weight because economic size represents the overall scale of productive economic activity within the country.

Natural resources were assigned a smaller but still significant weight because resource wealth can represent valuable economic assets and productive potential.

The model is intentionally simple and transparent. The objective was to construct a reproducible quantitative framework rather than create an overly complex model with assumptions that could not be defended.

The Python implementation normalized both variables and calculated the final weighted score directly.

22. COUNTRY VALUATION RESULTS

The final results were:

Rank	Country	GDP Score	Resource Score	Final Valuation Score
1	United States	100.00	5.01	71.50
2	China	76.72	6.68	55.71
3	Saudi Arabia	4.14	100.00	32.90

4	India	13.35	12.36	13.05
5	Singapore	1.86	0.00	1.30

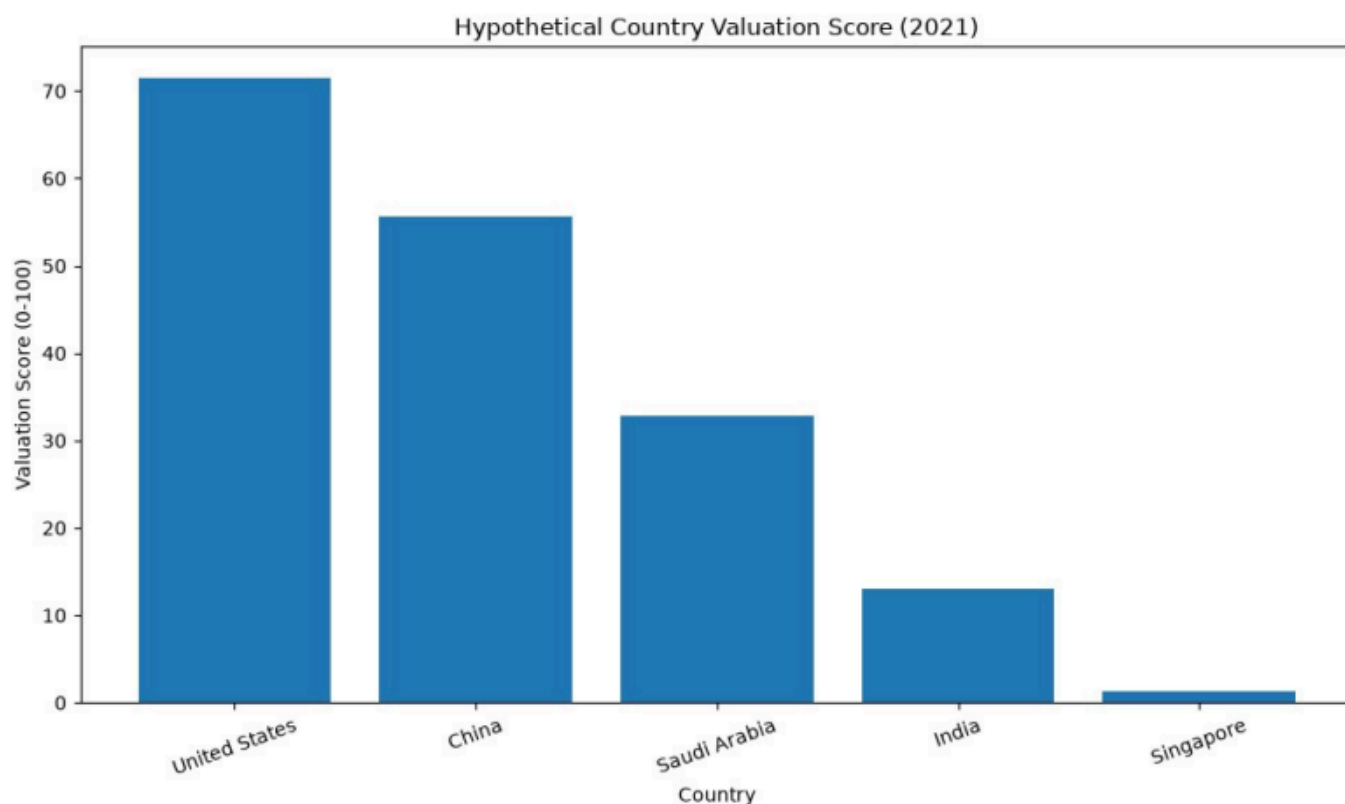


Figure 6: Hypothetical Country Valuation Score (2021)

23. INTERPRETATION OF RESULTS

23.1 United States – Rank 1

The United States achieved the highest hypothetical valuation score of 71.50.

The primary reason was its position as the largest economy in the selected sample. It received a GDP Score of 100.

Although its Natural Resource Score was relatively low compared with Saudi Arabia, the 70% weight assigned to GDP had a significant impact on the final score.

This result shows that under the selected framework, economic size was the most important driver of hypothetical country value.

23.2 China – Rank 2

China ranked second with a Country Valuation Score of 55.71.

China had a GDP Score of 76.72, reflecting its position as the second-largest economy among the selected countries.

Its Natural Resource Score was relatively low at 6.68. However, the large GDP component was sufficient to produce a strong overall valuation score.

This demonstrates the importance of the GDP weighting in the model.

23.3 Saudi Arabia – Rank 3

Saudi Arabia ranked third with a score of 32.90.

Saudi Arabia had the highest Natural Resource Score of 100 because it had the highest natural resource rents as a percentage of GDP among the selected countries.

However, its GDP Score was only 4.14 because its total economic size was substantially smaller than the United States and China.

The 70% GDP weighting therefore limited Saudi Arabia's final score despite its extremely strong resource profile.

This is a useful example of how the selected model weights influence the ranking.

23.4 India – Rank 4

India received a Country Valuation Score of 13.05.

India had a GDP Score of 13.35 and a Resource Score of 12.36.

Although India has a large economy, its GDP was significantly smaller than the United States and China in 2021. Its natural resource score was also much lower than Saudi Arabia's.

Under the specific two-factor framework used in this study, these values resulted in a relatively low final score.

This result should not be interpreted as an absolute measure of India's economic potential. It reflects only the two variables and weights selected for this particular model.

23.5 Singapore – Rank 5

Singapore received the lowest valuation score of 1.30.

Its GDP Score was 1.86, while its Natural Resource Score was approximately zero.

The result is mainly caused by the narrow structure of the model.

Singapore has important economic strengths, including human capital, financial services, infrastructure, institutional quality, and strategic economic importance. However, these variables were not included in the current two-factor model.

Therefore, Singapore's low score demonstrates one of the main limitations of the model: a country's value cannot realistically be represented completely using only GDP and natural resource rents.

24. QUESTION 2 CONCLUSION

The hypothetical country valuation framework successfully produced a transparent and reproducible numerical ranking.

The results were:

1. United States – 71.50
2. China – 55.71
3. Saudi Arabia – 32.90
4. India – 13.05
5. Singapore – 1.30

The United States ranked first because of its extremely large GDP. China ranked second because of its large economic size. Saudi Arabia ranked third because its very high natural resource value partially compensated for its smaller GDP.

The results demonstrate that the hypothetical value of a country depends heavily on the variables selected and the weights assigned to them.

Therefore, the final ranking should not be interpreted as an absolute ranking of the real-world value of countries.

It should instead be interpreted as:

A quantitative comparison of the selected countries under a specific hypothetical valuation framework using economic size and natural resource wealth.

The model could be improved in future research by including:

- Human capital
 - Infrastructure
 - Productivity
 - Government debt
 - Fiscal strength
 - Institutional quality
 - Country risk
 - Population demographics
 - Technological capability
 - Strategic location
 - Future economic growth
 - Environmental resources
-

25. OVERALL DISCUSSION

The two parts of this case study demonstrate that quantitative analysis requires more than simply generating a result.

In Question 1, the initial Gold-Silver pair had a strong correlation, but statistical testing did not provide sufficient evidence of cointegration. Instead of ignoring the result, the analysis moved to another economically related pair, XLE and VDE.

The XLE-VDE pair demonstrated statistically significant cointegration and stationarity of the constructed spread. The out-of-sample ADF test also supported the mean-reversion hypothesis.

However, the backtest showed that the strategy was not profitable over the selected period, particularly after transaction costs.

This demonstrates that:

Statistical significance does not guarantee investment profitability.

In Question 2, the country valuation model successfully produced a numerical ranking using real data. However, the results were strongly influenced by the selected variables and their weights.

This demonstrates that:

A quantitative valuation model is only as reliable as its assumptions, variables, and weighting methodology.

Both questions therefore highlight the importance of model validation and self-critique.

27. FINAL CONCLUSION

This case study applied quantitative methods to two different problems: statistical arbitrage and hypothetical country valuation.

In Question 1, the research began by analysing Gold and Silver. Although the pair had a strong correlation, the cointegration test did not provide sufficient evidence to support the pair as the final statistically validated portfolio. This result demonstrated why high correlation alone is not sufficient for pairs trading.

The analysis then identified XLE and VDE as a more suitable final portfolio because the two ETFs are economically related through their exposure to the energy sector.

The XLE-VDE pair showed evidence of cointegration and the constructed spread was found to be stationary during both the training and out-of-sample periods. The training ADF p-value was 0.029451 and the out-of-sample ADF p-value was 0.025275.

Despite this statistical evidence, the trading strategy generated negative returns, particularly after transaction costs.

Therefore, the final conclusion for Question 1 is:

The portfolio demonstrated statistical mean reversion, but the specific trading strategy was not sufficiently profitable to be considered practically investable under the assumptions used in the backtest.

In Question 2, a hypothetical country valuation model was developed using GDP and natural resource rents.

The model ranked the United States first, followed by China, Saudi Arabia, India, and Singapore.

The results demonstrated that a country's hypothetical value is highly dependent on the variables included in the model and the weights assigned to them.

Therefore, the final conclusion for Question 2 is:

The country valuation model provides a structured and transparent numerical framework for comparison, but it should not be interpreted as a literal or complete measure of a country's true economic value.

Overall, this case study demonstrates that good quantitative analysis requires not only building models but also questioning them.

The most important lessons from this project are:

1. High correlation does not prove a valid statistical arbitrage opportunity.
2. Economic reasoning should support statistical testing.

3. Out-of-sample validation is essential.
4. Statistical significance does not guarantee profitability.
5. Transaction costs can materially change backtest results.
6. Quantitative models depend on assumptions and selected variables.
7. Every model should be accompanied by an honest discussion of its limitations.

The project therefore concludes that rigorous quantitative analysis should combine **data, statistics, economic reasoning, realistic implementation, and self-critique** before a model is considered useful for decision-making.

28. TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used in this case study:

Python

Python was used for:

- Data cleaning
- Data manipulation
- Statistical testing
- Regression analysis
- Cointegration testing
- ADF testing
- Spread construction
- Z-score calculation
- Backtesting
- Performance calculation
- Country valuation scoring
- Data visualization

Libraries Used

- Pandas
- NumPy
- Matplotlib
- Statsmodels
- yfinance
- Seaborn

Tableau

Tableau was used to create visual dashboards for the Gold and Silver market analysis, including:

- Price trends
- Cumulative returns
- ₹1 Lakh investment growth
- Annual returns
- Drawdowns

Data Sources

- Yahoo Finance for financial market data
 - World Bank datasets for GDP and natural resource data
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29. REFERENCES

1. Yahoo Finance – Historical market data used for ETF and commodity-related analysis.
 2. World Bank – GDP data.
 3. World Bank – Total Natural Resources Rents (% of GDP) data.
 4. Python Pandas Documentation.
 5. NumPy Documentation.
 6. Statsmodels Documentation.
 7. Tableau – Data visualization and dashboard development.
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<https://drive.google.com/drive/folders/19QsBfTYreG8TYhiq8ADTlzdQw7oHiOWQ?usp=sharing>